a. The cube code (5 min)

0–1.5 min / 1 The cube code $[\![8,3,2]\!]$ Set the notation once: words are functions on the cube, CSS states are coset superpositions. Step 3: Z000 Z100 is a weight-two logical.

1.5–3 min / 2 A transversal T/T† layer gives logical CCZ on the cube W equals the distance to p minus four. Both words of a logical pair get the same phase.

3–4 min / 3 Weights of sums expand into overlaps General formula only. Pointwise check gives signed weights too. Terms with four or more words vanish mod 8.

4–5 min / 4 The gate condition is a congruence on generator rows Coefficients 1, minus 2, 4 give the mod 8, mod 4 and mod 2 conditions. The cube satisfies them.

b. History and the CH conditions (5 min)

5–7 min / 5 From magic-state distillation to triorthogonal codes Cite the CSS-T line of Camps-Moreno, López, Matthews et al. Nezami-Haah n+k at most 38 is not our bound.

7–8.5 min / 6 A triorthogonal matrix and a triorthogonal space differ Odd rows break the space definition. CCZ needs an odd logical triple.

8.5–10 min / 7 The nine CH conditions Mod 2 reduction of the row congruences. Quasi-transversal means up to a diagonal Clifford correction.

c. Results and finite geometry (10 min)

10–12 min / 8 Result: a 48-qubit code with a transversal T/T† CCZ State the result now that every term is defined. 39 to 46 excluded by Wills, Jain and Singh.

12–14 min / 9 Fix S using finite/projective geometry code Four steps. Only the logical rows are searched.

14–15.5 min / 10 Projectivity rules out weight-one and weight-two Z logicals Parity-check argument. Covers d_Z only.

15.5–17.5 min / 11 The support is projective space minus a subspace 63 points, remove the 15 with w = 00, three blocks of 16. This picture stays for the rest of the construction.

17.5–20 min / 12 Hyperplane sections give the two nonzero weights u1 cuts each block in half: 24. w1 takes two blocks: 32. Both divisible by 8.

d. Indicators, Reed–Muller codes, SAT (10 min)

20–21.5 min / 13 Column indicators record the generator matrix Cube: chi_G = y. Q48: chi_S = w1 + w2 + w1 w2.

21.5–23.5 min / 14 The CH conditions are inner products with monomials Overlap parity equals an inner product with a monomial. Coset of RM(r-4, r).

23.5–25.5 min / 15 The label map turns logical rows into Boolean unknowns chi_G factors through the labels. Step 2 shows the support of K1.

25.5–27 min / 16 SAT assigns three logical bits to each geometric point 144 label bits, 88 parity constraints, no distance constraints.

27–28.5 min / 17 The signs solve a linear system over $\mathbb{Z}_4$ Binary unknowns, linear over Z4. Pattern 26 T plus 22 T-dagger, checked on 512 codewords.

28.5–30 min / 18 A three-point witness makes the Z-distance exact Syndromes cancel, label sum 101 is nonzero.

e. Bounds and implications (10 min)

30–32 min / 19 Stabilizer supports are Reed–Muller words Indicator method from Nezami-Haah. Q48 gives a word of RM(2,6).

32–34 min / 20 Our exclusion argument covers every length through 38 Analytic reductions plus a complete finite enumeration.

34–36 min / 21 Concurrent classification closes the 39–46 gap CNOT plus S equivalence. No uniqueness claim at 47.

36–37.5 min / 22 What the 47-qubit comparison establishes The Jacinto code has no sign vector. This concerns that code only.

37.5–38.5 min / 23 Code length and distillation footprint are different resources n is not the only cost. No resource claim for Q48.

38.5–39.5 min / 24 The geometric construction produces a family p = 4, q = 2 gives 48. Lower bounds on counts except the 48 census.

39.5–40 min / 25 Open questions connect algebra to implementation Leave the QR codes up during questions.

40 minutes: finish and invite questions.